

Cheat Sheet: Plotly and Dash

1 Plotly Express Functions

Function	Description	Syntax	Example
scatter	Create a scatter plot	<code>px.scatter(dataframe, x=x_column, y=y_column)</code>	<code>px.scatter(df, x=age_array, y=income_array)</code>
line	Create a line plot	<code>px.line(x=x_column, y=y_column, title='title')</code>	<code>px.line(x=months_array, y=no_bicycle_sold_array)</code>
bar	Create a bar plot	<code>px.bar(x=x_column, y=y_column, title='title')</code>	<code>px.bar(x=grade_array, y=score_array, title='Pass Percentage')</code>
sunburst	Create a sunburst plot	<code>px.sunburst(dataframe, path=[col1, col2, ...], values='column', title='title')</code>	<code>px.sunburst(data, path=['Month', 'DestStateName'], values='Flights', title='Flight Distribution Hierarchy')</code>
histogram	Create a histogram	<code>px.histogram(x=x, title='title')</code>	<code>px.histogram(x=heights_array, title='Distribution of Heights')</code>
bubble	Create a bubble chart	<code>px.scatter(dataframe, x=x, y=y, size=size, title='title')</code>	<code>px.scatter(bub_data, x='City', y='Numberofcrimes', size='Numberofcrimes', hover_name='City', title='Crime Statistics')</code>
pie	Create a pie chart	<code>px.pie(values=x, names=y, title='title')</code>	<code>px.pie(values=exp_percent, names=house_holdcategories, title='Household Expenditure')</code>

2 Plotly Graph Objects

Function	Description	Syntax	Example
go.Scatter (Scatter Plot)	Create a scatter plot	<code>go.Scatter(x=x, y=y, mode='markers')</code>	<code>go.Scatter(x=age_array, y=income_array, mode='markers')</code>
go.Scatter (Line Plot)	Create a line plot	<code>go.Scatter(x=x, y=y, mode='lines')</code>	<code>go.Scatter(x=months_array, y=no_bicycle_sold_array, mode='lines')</code>
add_trace	Add additional traces to an existing figure	<code>fig.add_trace(trace_object)</code>	<code>fig.add_trace(go.Scatter(x=months_array, y=no_bicycle_sold_array))</code>
update_layout	Update figure layout such as title, axis labels, and annotations	<code>fig.update_layout(layout_object)</code>	<code>fig.update_layout(title='Bicycle Sales', xaxis_title='Months', yaxis_title='Number of Bicycles Sold')</code>

3 Dash Components

Component	Description	Syntax	Example
dash_core_components.Input (<code>dcc.Input</code>)	Create an input component	<code>dcc.Input(value='', type='text')</code>	<code>dcc.Input(value='Hello', type='text')</code>
dash_core_components.Graph (<code>dcc.Graph</code>)	Create a graph component	<code>dcc.Graph(figure=fig)</code>	<code>dcc.Graph(figure=fig)</code>
dash_html_components.Div (<code>html.Div</code>)	Create a div element	<code>html.Div(children=component_list)</code>	<code>html.Div(children=[html.H1('Hello Dash'), html.P('Welcome to Dash')])</code>
dash_core_components.Dropdown (<code>dcc.Dropdown</code>)	Create a dropdown component	<code>dcc.Dropdown(options=options_list, value=default_value)</code>	<code>dcc.Dropdown(options=[{'label': 'Option 1', 'value': '1'}, {'label': 'Option 2', 'value': '2'}], value='1')</code>

4 Quick Reference

4.1 Plotly Express Charts

- `px.scatter()` → Scatter Plot
- `px.line()` → Line Plot
- `px.bar()` → Bar Plot
- `px.sunburst()` → Sunburst Plot
- `px.histogram()` → Histogram
- `px.scatter(..., size=...)` → Bubble Chart
- `px.pie()` → Pie Chart

4.2 Plotly Graph Objects

- `go.Scatter(mode='markers')` → Scatter Plot
- `go.Scatter(mode='lines')` → Line Plot
- `fig.add_trace()` → Add a new trace to a figure
- `fig.update_layout()` → Update chart layout and labels

4.3 Dash Components

- `dcc.Input()` → User input field
- `dcc.Graph()` → Display Plotly figures
- `html.Div()` → Container for Dash components
- `dcc.Dropdown()` → Dropdown selection menu